Tackling CASMI 2022: Solutions from Mol3DNet and MetaboSLASH

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Abstract

Critical Assessment of Small Molecule Identification (CASMI) in 2022 aims to identify small molecules from the increasing tandem mass (MS/MS) spectrometry database. In the contest, raw LC-accurate mass MS/MS data in both positive and negative mode and a list of unidentified small molecules are provided. A deep learning model, Mol3DNet, is trained to predict MS/MS from molecular 3D conformers. The training data is from the public libraries, including GNPS, NIST20, and MoNA. Then we established a predicted MS/MS library from 216,704 compounds of Human Metabolome Database (HMDB), which contains the spectra in different collision energies and precursor types. With the predicted library, the fast and sensitive searching software, msSLASH [4], is applied to identify compounds with their experimental mass and MS/MS. In the end, the compounds with SMILES can be classified by ClassyFire [1].

1 Methods

1.1 Establish predicted MS/MS library by Mol3DNet

Mol3DNet is a deep learning model predicting accurate MS/MS from molecular 3D conformers. The inputs of this model is the SMILES of compounds and other experimental conditions, including collision energy and precursor type. The SMILES is convert into 3D conformers by `Chem.MolFromSmiles()` and `AllChem.EmbedMolecule()` in RDKit library [3], which is then encoded into points of set. The spectra is binned into a fix-length vector, and intensities are normalized into from 0 to 1. In Mol3DNet, a convolutional operation, MolConv, is designed to extract features from conformers. With the sequential MolConv, the 3D compounds are embedded into a vector in latent space. Then multilayer perceptron (MLP) with residual connections decodes the vector to MS/MS. Cosine distance (equation 1) is used as loss function and metrics.

$$\text{cosine distance} = 1 - \text{cosine similarity} = 1 - \frac{A \cdot B}{||A||||B||} \tag{1}$$

where A and B are two spectra vectors.

Train Mol3DNet: Mol3DNet is trained on public libraries, including spectra from Global Natural Products Social Molecular Networking (GNPS) [5], NIST20 [7], and MassBank of North America (MoNA) [2]. We filter out spectra from Higher-energy collisional dissociation (HCD) spectra (acquired using the ion trap MS instruments) and the combination of Q-TOF spectra (acquired using the quadrupole/time-of-flight MS instruments) and QqQ spectra (acquired using the triple-quadrupole MS instruments) as training datasets.

Establish predicted library: The metabolite structures in `sdf` format without spectra can be downloaded from HMDB [6] database, which are the input of Mol3DNet when inference. Because there are tons of experimental conditions, we choose the common ones:

- 3 level collision energy: 10 eV, 20 eV, 40 eV
- 8 precursor types: 'M+H', 'M-H', 'M+H-H2O', 'M+Na', 'M+H-NH3', 'M+H-2H2O', 'M-H-H2O', 'M+Cl', 'M+OH'

To handle the large quantity data, all the SMILES, compound mass, retention time, collision energy, precursor type and predicted mass spectra are saved in a series of `mgf` format files, which can be used as a reference library in the searching algorithm.

1.2 Identify compounds by msSLASH

The msSLASH [4] software implements a fast spectral library searching algorithm based on the Locality-Sensitive Hashing (LSH) technique. The reference library and query spectra are first converted into bit-string using LSH functions. Then the similarity between the spectra with highly similar bit-string can be computed.

Search target MS/MS: We first process the raw MS/MS data into `mgf` format, including compound mass and retention time. The codes for msSLASH is available here: https://github.com/COL-IU/msSLASH. The setting parameters we used is shown in table 1.2. In addition to the predicted library, we also searching the spectra in assembled public libraries including GNPS, NIST20, MoNA, and labeled HMDB. The spectra with max cosine similarity among libraries is picked as final results.

Setting Parameter Description	Value
Minimize m/z for peaks to consider	10
Fragment precision in Da	0.2
Precursor mass tolerance in Da	2
Peak intensity rescale method	sqrt transformation
Hash functions per hash table	10
Iteration for searching with LSH	200

Table 1: The settings of msSLASH.

Classify compounds: The hierarchical chemical classification of compounds is calculated at the website of ClassyFire. The classification results, including *kingdom*, *superclass*, *class*, and *subclass*, are organized in `json` format. To make the classification in a higher-order we report the *class* as the compound class.

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